

CUSTOMER CHURN ANALYSIS

Project Submitted By

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Internship

Data Science & Analytics Internship – Enginow

Project Title

Customer Churn Analysis Using Data Analytics

Executive Summary

Customer retention is one of the most critical factors influencing business growth and profitability. Acquiring new customers is significantly more expensive than retaining existing ones. Therefore, understanding the reasons behind customer churn is essential for organizations seeking sustainable growth.

This project focuses on predicting customer churn using Data Analytics techniques and identifying the key factors that influence customer retention. The analysis was conducted using a telecom customer dataset containing demographic information, service details, contract information, payment methods, and customer billing data.

The project involved data preprocessing, exploratory data analysis (EDA), feature engineering, Data Analytics model development, and business insight generation. The findings provide actionable recommendations that organizations can implement to reduce churn and improve customer loyalty.

1. Introduction

Customer churn refers to the phenomenon where customers discontinue using a company's products or services. High churn rates directly impact revenue, profitability, and customer acquisition costs.

The primary objective of this project is to identify customers who are likely to churn and understand the factors contributing to their decision. By leveraging Data Analytics algorithms and data analytics techniques, organizations can proactively target at-risk customers and improve retention strategies.

2. Project Objectives

The major objectives of this project are:

- Analyze customer behavior and churn patterns.
 - Identify key factors influencing customer churn.
 - Perform data preprocessing and exploratory analysis.
 - Build Data Analytics classification models.
 - Evaluate model performance using standard metrics.
 - Generate actionable business recommendations.
 - Assist organizations in reducing customer attrition.
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3. Dataset Description

The dataset used in this project contains customer information collected from a telecommunications company.

Dataset Features

Feature	Description
customerID	Unique customer identifier
gender	Customer gender
SeniorCitizen	Senior citizen status

Partner	Whether customer has a partner
Dependents	Whether customer has dependents
tenure	Number of months customer stayed
PhoneService	Phone service subscription
MultipleLines	Multiple phone lines
InternetService	Type of internet service
OnlineSecurity	Online security subscription
OnlineBackup	Online backup subscription
DeviceProtection	Device protection service
TechSupport	Technical support subscription
StreamingTV	TV streaming service
StreamingMovies	Movie streaming service
Contract	Contract type
PaperlessBilling	Billing preference
PaymentMethod	Payment method
MonthlyCharges	Monthly bill amount
TotalCharges	Total charges paid
Churn	Target variable

Dataset Size:

- Records: 7043
- Features: 21

4. Data Preprocessing

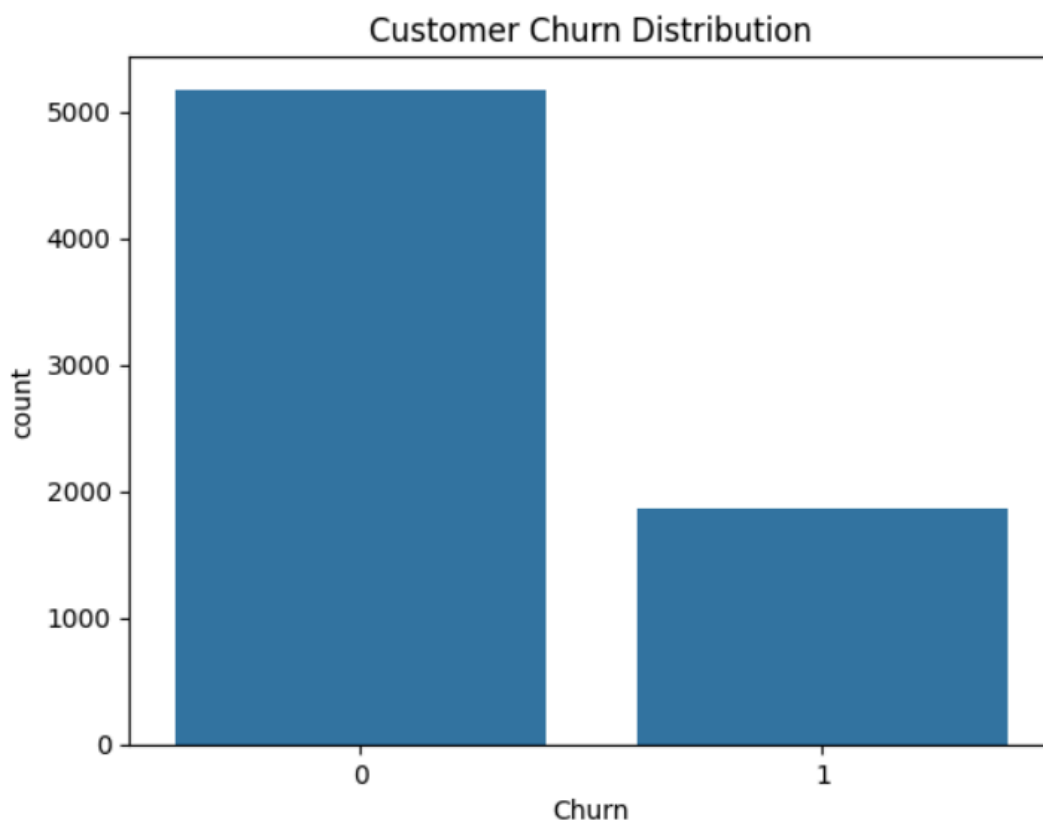
Data preprocessing is a crucial step in Data Analytics projects as it improves data quality and model performance. The following preprocessing tasks were performed:

- **Missing Value Handling:** The dataset was inspected for missing values. Missing entries in the `TotalCharges` column were identified and appropriately handled.
 - **Data Type Conversion:** The `TotalCharges` feature was converted from object datatype to numerical format to facilitate analysis and model training.
 - **Categorical Encoding:** Categorical variables such as `Contract` , `PaymentMethod` , and `InternetService` were transformed into numerical values using encoding techniques.
 - **Duplicate Removal:** Duplicate records were checked and removed where necessary to maintain data integrity.
 - **Feature Selection:** The target variable `Churn` was separated from predictor variables for Data Analytics model development.
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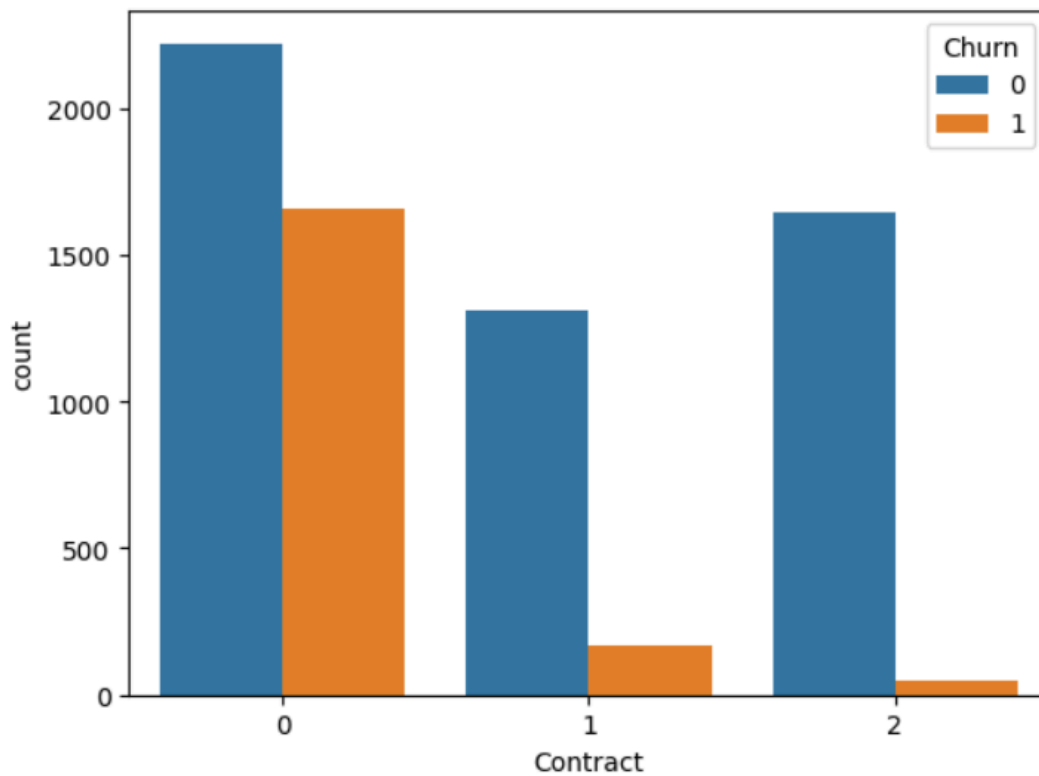
5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand customer behavior and identify patterns associated with churn.

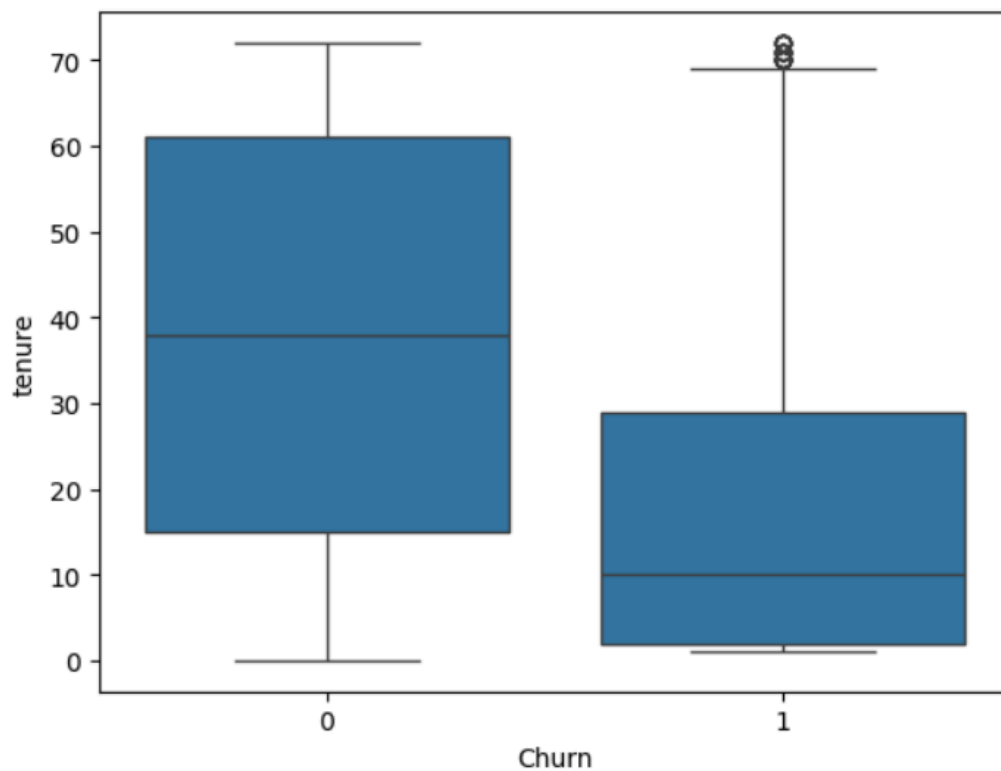
- **Churn Distribution:** The analysis revealed that a majority of customers remain with the company, while a smaller percentage discontinue services.



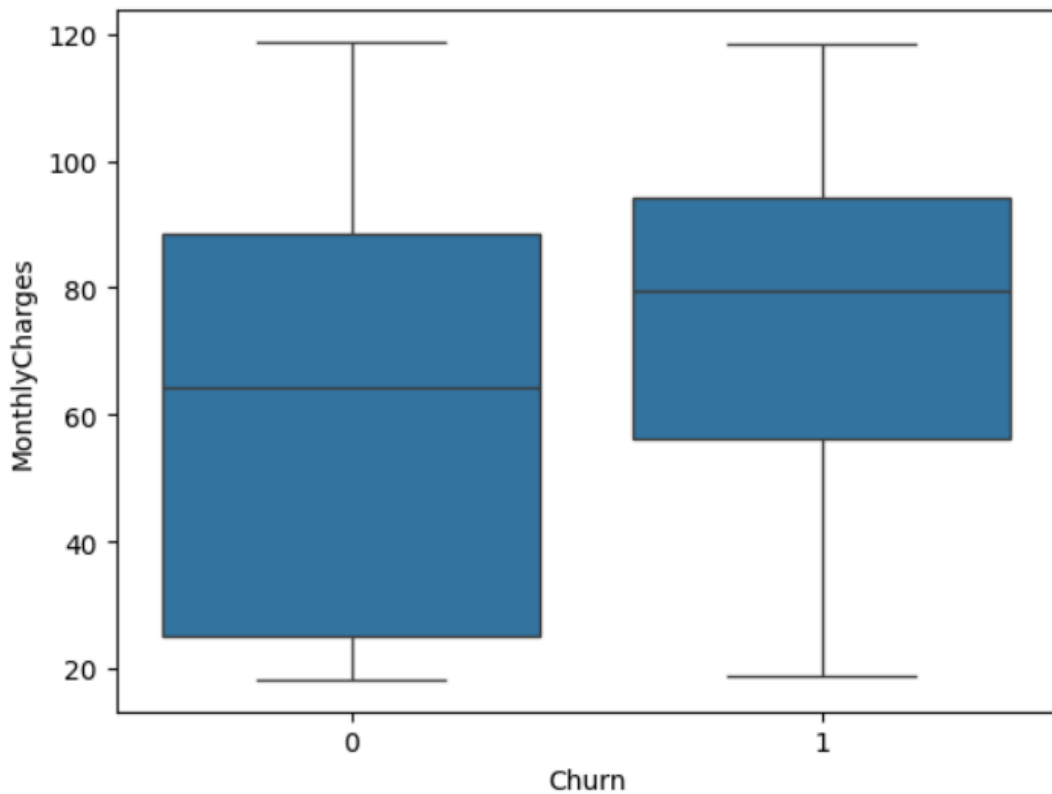
- **Contract Type vs Churn:** Customers with month-to-month contracts exhibited substantially higher churn rates than customers with annual or biannual contracts.



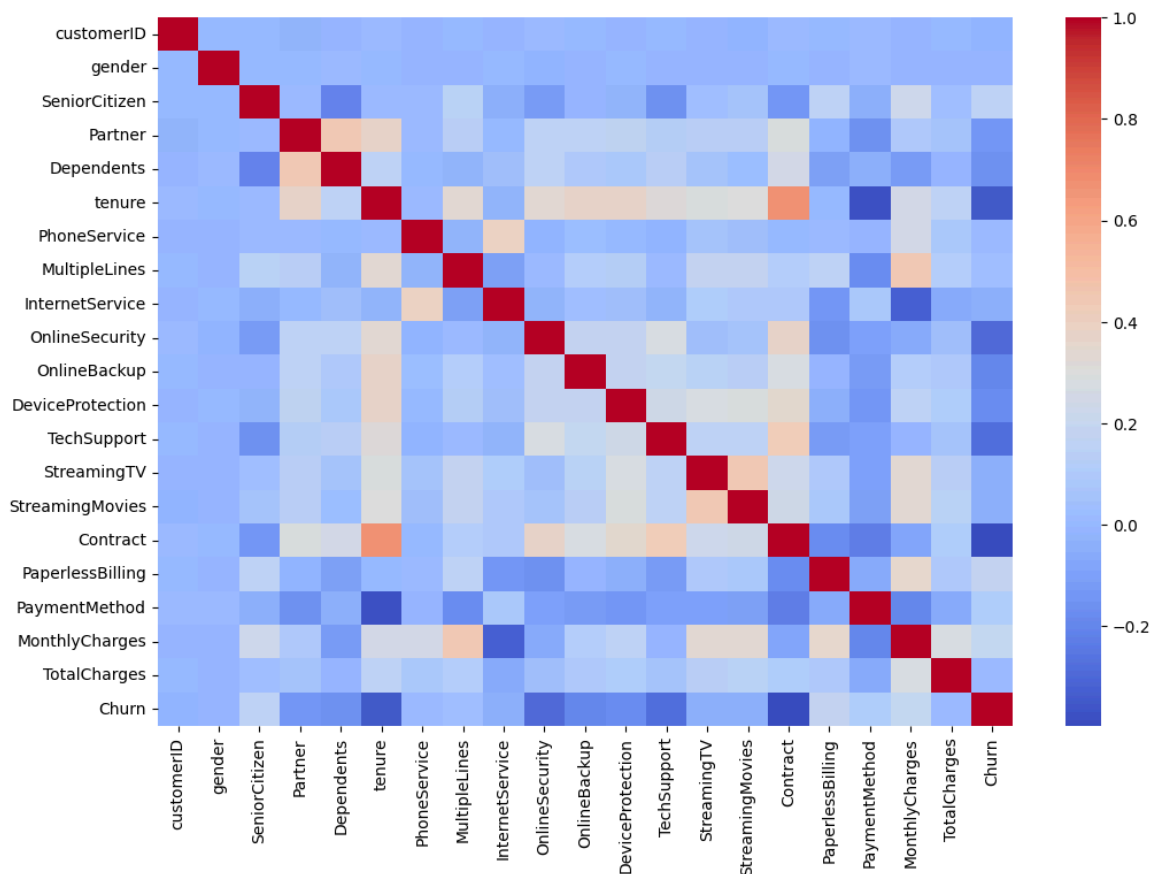
- **Tenure Analysis:** Customers with shorter tenure demonstrated a greater likelihood of churn compared to long-term customers.



- **Monthly Charges Analysis:** Customers with higher monthly charges showed increased churn probability.



- **Correlation Analysis:** Correlation analysis was performed to identify relationships between variables and their impact on churn behavior.



6. Data Analytics Model Development

Three Data Analytics classification algorithms were implemented:

Logistic Regression

Logistic Regression was used as a baseline classification model due to its simplicity and interpretability.

- **Advantages:** Easy to interpret, fast training time, effective for binary classification.

Decision Tree Classifier

Decision Trees were used to capture non-linear relationships within the dataset.

- **Advantages:** Easy visualization, handles complex relationships, requires minimal preprocessing.

Random Forest Classifier

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting.

- **Advantages:** High accuracy, robust performance, effective feature importance estimation.
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7. Model Evaluation

The models were evaluated using Accuracy, Precision, Recall, F1 Score, and a Confusion Matrix.

Model Comparison

Model	Accuracy
Logistic Regression	0.813342
Decision Tree	0.800568
Random Forest	0.717530

Based on the evaluation results, the Random Forest model demonstrated superior performance and was selected as the final predictive model.

8. Key Business Insights

- **Insight 1: Month-to-Month Customers Exhibit the Highest Churn Risk**
Customers with month-to-month contracts have the highest churn rates because they can easily discontinue services without long-term commitments. This customer segment should be prioritized for retention efforts.
- **Insight 2: Low-Tenure Customers Are More Likely to Leave**
Customers in the early stages of their relationship with the company are more likely to churn. Effective onboarding and early customer engagement are critical.
- **Insight 3: Higher Monthly Charges Increase Churn Probability**
Customers paying higher monthly fees are more likely to evaluate alternative providers. Pricing strategies should be reviewed for high-cost segments.

- **Insight 4: Service Bundling Improves Retention**

Customers utilizing multiple services tend to remain loyal for longer periods, indicating the effectiveness of bundled offerings.

- **Insight 5: Long-Term Contracts Improve Customer Loyalty**

Customers enrolled in annual or multi-year contracts exhibit significantly lower churn rates and contribute to stable recurring revenue.

9. Business Recommendations

- **Recommendation 1: Contract Migration Incentives**

Encourage customers to switch from month-to-month contracts to annual plans through discounts and promotional offers.

- **Recommendation 2: Structured Customer Onboarding**

Develop a structured onboarding program for newly acquired customers to improve engagement during the first few months.

- **Recommendation 3: Personalized Pricing Models**

Introduce personalized pricing strategies for customers with higher monthly charges.

- **Recommendation 4: Proactive Risk Monitoring**

Implement predictive churn monitoring systems to identify high-risk customers before they leave.

- **Recommendation 5: Long-Term Loyalty Rewards**

Offer loyalty rewards, service upgrades, and exclusive benefits to long-term customers.

- **Recommendation 6: Strategic Service Bundling**

Promote bundled service packages to increase customer dependency on company services and reduce churn.

10. Conclusion

Customer churn prediction plays a vital role in helping organizations retain customers and improve profitability. Through exploratory data analysis and Data Analytics techniques, significant factors influencing churn were identified.

The project demonstrated that contract type, tenure, monthly charges, and customer engagement levels are among the most important drivers of churn behavior. Among the evaluated models, Random Forest provided the strongest predictive performance.

By implementing the recommendations outlined in this report, organizations can proactively reduce churn, improve customer satisfaction, and achieve sustainable business growth.

11. Future Scope

Future improvements to this framework may include:

- Hyperparameter tuning for higher model accuracy.
 - Deployment of the churn prediction model as a web application.
 - Real-time churn monitoring dashboards.
 - Advanced ensemble learning techniques.
 - Integration with CRM systems for automated retention campaigns.
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References

1. IBM Telco Customer Churn Dataset
2. Scikit-Learn Documentation
3. Pandas Documentation
4. Data Analytics for Business Analytics Resources